

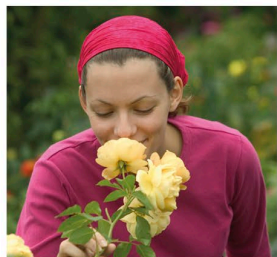
INFORMATION PROCESSING

THE BODY IS A DYNAMIC MECHANISM WHOSE INTERACTING PARTS REQUIRE CONTROL AND COORDINATION. TWO BODY SYSTEMS ARE RESPONSIBLE FOR THESE INFORMATION-PROCESSING FUNCTIONS: THE NERVOUS AND ENDOCRINE SYSTEMS.

Information processing involves inputs, evaluation, and decision-making, followed by outputs. The body receives inputs from the senses. The brain is the central processing unit whose outputs control the physical actions of muscles and chemical responses of glands. Both nerves and hormones are involved in data management.

ELECTRICAL AND CHEMICAL PATHWAYS

The “language” of the nervous system is tiny electrical impulses. Every second, millions pass through the body’s nerve network, conveying information to and from the brain. Information from the senses flows to the brain, where it is analysed. Decisions are reached, and command messages – also in the form of electrical impulses – travel along motor nerves to the muscles to stimulate and coordinate their contractions. In addition, microreceptors monitor conditions inside the body and feed data about it to the unconscious part of the brain, which automatically evaluates the data and sends out impulses to various parts of the body to keep the internal environment at the optimum for body functioning.



SELECTIVE FOCUS

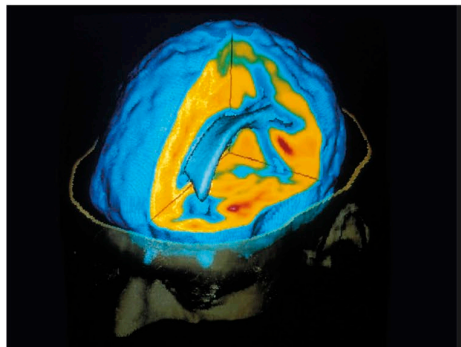
The nose sends streams of “smell” nerve signals to the brain. We can choose to ignore these or to focus on them, as part of the mind’s selective awareness.

NERVOUS SYSTEM

76–129

ENDOCRINE SYSTEM

130–143



BRAIN ACTIVITY

This image is a three-dimensional functional MRI scan showing brain activity during speech. Red indicates areas of high activity, yellow indicates medium activity, while blue indicates low activity.

Different information carriers called hormones are secreted by endocrine glands into the bloodstream to stimulate distant tissues to action. More than 50 hormones circulate in the bloodstream. The specific molecular structure of each hormone stimulates only those cells that have suitable receptors on their surface, instructing the cells to carry out certain processes. In general, nerves work fast – within fractions of a second. Most hormones function over longer time periods – minutes, days, or even months. Long-lasting effects, as in growth hormone for example, occur because the hormone is continuously secreted over a period of many years; an individual dose would last only a few days.